The Basics of C Programming

Beginner's Guide

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1 Introduction

C is a powerful, efficient, and widely used programming language. It is the foundation for many modern languages like C++, Java, and Python. This document outlines the core concepts required to get started.

2 The Structure of a C Program

Every C program requires a library to handle input/output and an entry point called `main`.

```c
#include <stdio.h> // 1. Preprocessor Command

int main() {        // 2. The Main Function
    // 3. The Statement (prints to console)
    printf("Hello, World!\n");

    return 0;       // 4. Return Value (Success)
}
```

Listing 1: Hello World Example

3 Variables and Data Types

In C, you must declare the type of data a variable will hold.

Type	Description	Example
`int`	Integers (Whole numbers)	`int age = 25;`
`float`	Decimal numbers	`float price = 10.99;`
`double`	High precision decimals	`double pi = 3.14159;`
`char`	Single characters	`char grade = 'A';`

4 Control Flow

Logic is handled through conditional statements and loops.

4.1 If / Else Statements

```c
int age = 18;

if (age >= 18) {
    printf("You are an adult.");
} else {
    printf("You are a minor.");
}
```

4.2 Loops

Use a **For Loop** when you know the number of iterations:

```
for (int i = 0; i < 5; i++) {
    printf("%d ", i); // Output: 0 1 2 3 4
}
```

5 Functions

Functions allow you to compartmentalize code.

```
// Function Declaration
int add(int a, int b) {
    return a + b;
}

int main() {
    int sum = add(5, 10);
    printf("The sum is: %d", sum);
    return 0;
}
```

6 Pointers

Pointers are variables that store the **memory address** of another variable. They are crucial for efficient memory management in C.

```
int myVar = 10;
int *ptr = &myVar; // ptr holds the address of myVar

printf("%p", ptr); // Prints the memory address
```

7 Summary Checklist

- **Compile:** C must be compiled (e.g., using GCC) before it runs.
- **Semicolons:** Every statement must end with a semicolon (;).
- **Entry Point:** Execution always begins at `main()`.